The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 August 2023.

INCH-POUND

MIL-PRF-19500/772A
w/AMENDMENT 1
14 April 2023
SUPERSEDING
MIL-PRF-19500/772A
15 July 2014

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, UNIPOLAR TRANSIENT VOLTAGE SUPPRESSOR, TYPES 1N8036 THROUGH 1N8072, 1N8073 THROUGH 1N8109, 1N8110 THROUGH 1N8146, THROUGH-HOLE AND SURFACE MOUNT PACKAGES, QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 150 watt, 500 watt and 1,500 watt peak pulse power, unipolar silicon transient voltage suppressor diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. The device package outlines are as follows: axial leaded through hole package in accordance with figure 1 and square end-cap surface mount US package in accordance with figure 2.

1.3 Maximum ratings. Maximum ratings are as shown in columns 4, 6, and 7 of the electrical characteristics table herein and as follows:

a. $P_{M(AV)}$ = 1 W (for 150 W peak pulse power devices), 2 W (for 500 W peak pulse power devices), and 3 W (for 1,500 W peak pulse power devices) at T_A = +25°C (see figures 3, 4, and 5 for derating).

b. P_{PP} = 150 W (1N8036 through 1N8072 (including US suffix versions)), P_{PP} = 500 W (1N8073 through 1N8109 (including US suffix versions), and P_{PP} = 1,500 W (1N8110 through 1N8146 (including US suffix versions) at t_P = 1 ms (see figure 6).

c. $P_{M(AV)}$ = 1.5 W (for 150 W peak pulse power devices), P_R = 3 W (for 500 W peak pulse power devices) and 5 W (for 1,500 W peak pulse power devices) at T_L = +75°C for L = 0.375 inch (9.53 mm).

d. I_{FSM} = 25 amps (for 150 W peak pulse power devices), 80 amps (for 500 W peak pulse power devices), and 130 amps (for 1,500 W peak pulse power devices) at t_P = 8.33 ms (T_A = +25C).

e. -55°C ≤ T_J ≤ +175°C, -55°C ≤ T_{STG} ≤ +175°C (ambient temperatures).

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1.4 Primary electrical characteristics. Primary electrical characteristics are as shown in columns 2, 5, and 6 of the electrical characteristics table herein.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 Device type. The designation system for the device types of diodes covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The diode of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the diode covered by this specification sheet are as follows:

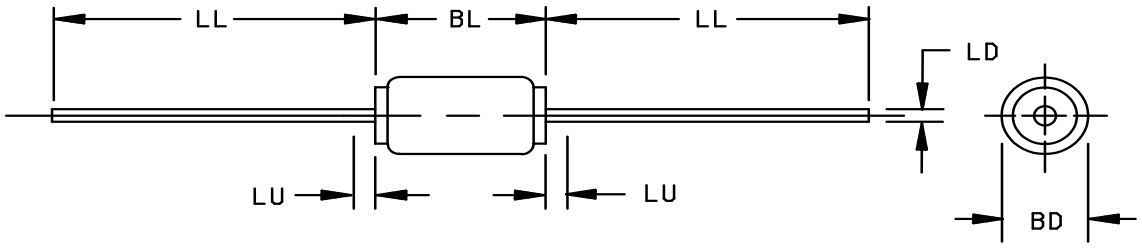
						8036	8037	8038	8039
8040	8041	8042	8043	8044	8045	8046	8047	8048	8049
8050	8051	8052	8053	8054	8055	8056	8057	8058	8059
8060	8061	8062	8063	8064	8065	8066	8067	8068	8069
8070	8071	8072	8073	8074	8075	8076	8077	8078	8079
8080	8081	8082	8083	8084	8085	8086	8087	8088	8089
8090	8091	8092	8093	8094	8095	8096	8097	8098	8099
8100	8101	8102	8103	8104	8105	8106	8107	8108	8109
8110	8111	8112	8113	8114	8115	8116	8117	8118	8119
8120	8121	8122	8123	8124	8125	8126	8127	8128	8129
8130	8131	8132	8133	8134	8135	8136	8137	8138	8139
8140	8141	8142	8143	8144	8145	8146			

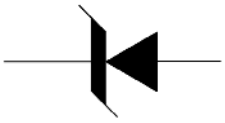
1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated into the PINs for this specification sheet.

	A blank package designators indicates a through hole mount, axial package (see figure 1).
US	Indicates a surface mount, square endcap package (see figure 2).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

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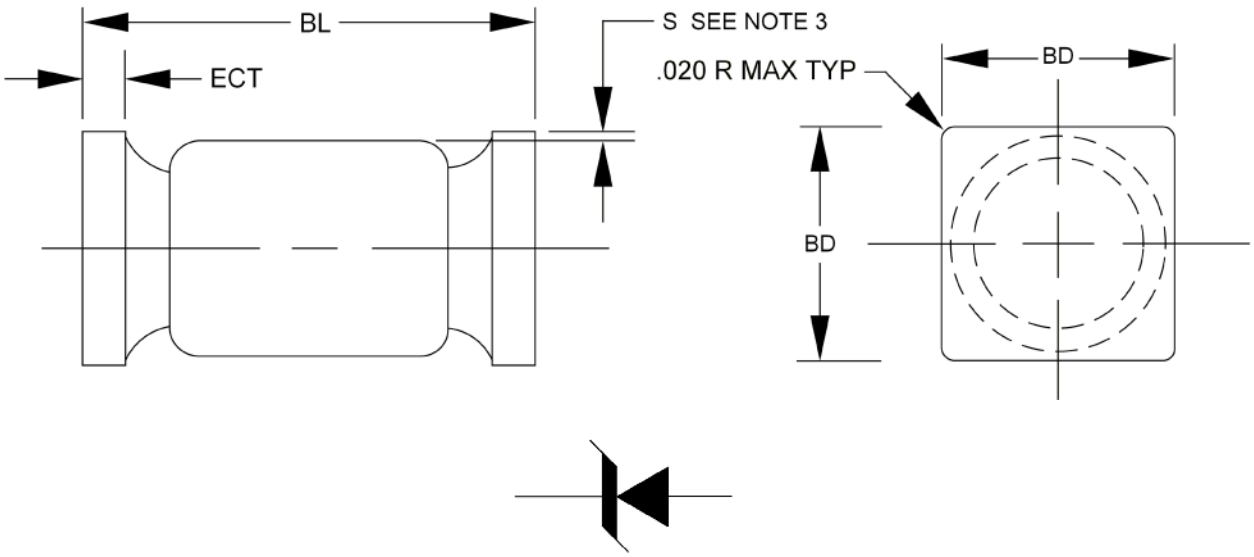
SCHEMATIC SYMBOL

	Dimensions												
	1N8036 through 1N8072				1N8073 through 1N8109				1N8110 through 1N8146				Note
Symbol	inch	inch	mm	mm	inch	inch	mm	mm	inch	inch	mm	mm	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
BD	.060	.085	1.52	2.16	.085	.140	2.16	3.56	.135	.185	3.43	4.70	2
BL	.106	.160	2.69	4.06	.140	.185	3.56	4.70	.140	.195	3.56	4.95	
LD	.028	.032	0.71	0.81	.036	.042	0.91	1.07	.036	.042	0.91	1.07	
LL	.800	1.300	20.32	33.02	0.900	1.30	22.86	33.02	0.900	1.30	22.86	33.02	4
LU		.050		1.27		.050		1.27		.050		1.27	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimension BD shall be measured at the largest diameter. Package contour optional within dimensions BD and length BL.
3. Dimension LU lead diameter uncontrolled in this area.

FIGURE 1. Physical dimensions of axial leaded package.

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SCHEMATIC SYMBOL

	Dimensions											
Symbol	1N8036US through 1N8072US				1N8073US through 1N8109US				1N8110US through 1N8146US			
	Inch	Inch	mm	mm	Inch	Inch	mm	mm	Inch	Inch	mm	mm
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
BD	.091	.103	2.31	2.62	.137	.148	3.48	3.76	.183	.202	4.65	5.13
BL	.168	.200	4.28	5.08	.200	.225	5.08	5.72	.205	.245	5.21	6.22
ECT	.019	.028	0.48	0.71	.019	.028	0.48	0.71	.019	.028	0.48	0.71
S	.003		0.08		.003		0.08		.003		0.08	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Minimum clearance of glass body to mounting surface on all orientations.

FIGURE 2. Physical dimensions of round body, square end-cap surface mount (US) package.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{(BR)}$	Reverse breakdown current at the specified condition.
$V_{C(max)}$	Clamping voltage. The maximum peak voltage appearing across the device when subjected to the peak pulse current I_{PP} in the reverse direction
V_{CF}	Forward clamping voltage. The maximum peak voltage appearing across the device when subjected to the peak pulse current I_{PP} in the forward direction.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 and 2 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.4.2 Metallurgical bond construction. Devices shall be metallurgically bonded, thermally matched, non-cavity, double-plug construction in accordance with the requirements of category I (see MIL-PRF-19500), and herein. The "US" version shall be structurally identical to the axial leaded type except for lead configuration.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. All device marking (including all marking which is omitted on the device), except for polarity and serial numbers, shall appear on the unit package used as the initial protection for delivery.

3.5.1 Marking of US devices. At the option of the manufacturer, US version devices may include laser marking on an end-cap, to include PIN and lot date code for all levels. JANS levels shall also include serialization.

3.5.2 Device marking abbreviations. The prefixes JAN, JANTX, JANTXV, or JANS may be abbreviated as J, JX, JV, or JS, respectively. (For example: The PIN JANS1N8036US may be reduced to JS8036).

3.5.3 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and 1.4, tables I and II.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in tables I and II.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4, tables I, II and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. A separate qualification shall be required for the 150-watt, 500-watt, and 1,500-watt peak pulse power device, respectively.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
5	Not applicable	Not applicable
9, 10, 11	Not applicable	Not applicable
12	See 4.5.1	See 4.5.1
13	Interim electrical, delta, and group A, subgroup 2, electrical parameters not applicable for this screen (performed in screen 12).	Interim electrical, delta, and group A, subgroup 2, electrical parameters not applicable for this screen (performed in screen 12).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. A separate quality conformance inspection shall be required for the 150-watt, 500-watt, and 1,500-watt peak pulse power devices, respectively.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

Subgroup	Method	Condition
B4		Not applicable.
B5	1027	T_A = +100°C minimum (see 4.5.2 and 4.5.3).

4.4.2.2 Quality levels JAN, JANTX, and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	See 4.5.1.
B5		Not applicable.
B6		Delta limits: $\Delta I_{D1} \leq$ 100 percent of initial reading or 20 percent of column 5 of table II, whichever is greater; $\Delta V_{(BR)1} \leq$ 5 percent of initial value.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows.

Subgroup	Method	Condition
C2	2036	Axial leaded devices only. The details for terminal strength testing shall be as follows: Tension (condition A). Applicable for all types using the following details: For 1N8036 through 1N8072; weight = 10 pounds (4.54 kg); t = 15 seconds minimum. For 1N8073 through 1N8109; weight = 12 pounds (5.44 kg); t = 15 seconds minimum. For 1N8110 through 1N8146; weight = 20 pounds (9.1 kg); t = 15 seconds minimum. Lead fatigue (condition E). Applicable for all axial leaded types.
C2	2038	US surface mount devices only. The details for end-cap bond integrity shall be as follows: Tab pull (condition B). Applicable for all types using the following details: For 1N8036 through 1N8072; weight = 10 pounds (4.54 kg), t = 15 seconds minimum. For 1N8073 through 1N8109; weight = 12 pounds (5.44 kg); t = 15 seconds minimum. For 1N8110 through 1N8146; weight = 20 pounds (9.1 kg); t = 15 seconds minimum.
C6	1026	See 4.5.1 and 4.5.3.
C7	4071	$I_{(BR)}$ = column 3 of table II, T_1 = +25°C ±3°C, T_2 = T_1 +100°C; sampling plan shall be 45 devices, c = 0; $\alpha_{V(BR)}$ = column 8 of table II.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Power burn-in and steady-state operation life test conditions. The test conditions and order of events shall be as follows:

a. Pulse in accordance with 4.5.3. Pulse 20 times for screening and group B and 100 times for group C at T_A = +25°C.

b. Read and record I_D and $V_{(BR)1}$ at T_A = +25°C, remove defective devices and record number of failures.

c. Apply the working peak voltage (V_{WM}) (column 4 of table II) at T_A = +125°C as follows:

(1) For 96 hours (JANTX and JANTXV) and 240 hours (JANS) for the screening test.

(2) For 340 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

(3) For 1000 hours for group C steady-state operation life test.

d. Read and record I_D at T_A = +25°C. Devices with ΔI_D > 50 percent (100 percent for steady-state operation life) of the initial reading or 20 percent of column 5 of table II, whichever is greater, shall be considered defective. Remove defective devices and record the number of failures.

4.5.1.1 Group C steady-state operation life test (alternate procedure). When the group B 340-hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.5.1 with the following exceptions: 4.5.1.d shall be repeated at the end of 1,000 hours.

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4.5.2 Accelerated steady-state operation life. This test shall be conducted with the devices subjected to the breakdown current specified in column 10 of table II in the reverse polarity for 96 +8, -4 hours. At the beginning of the test and at the end of the time period, the devices shall be temperature stabilized at T_A = +25°C and subjected to pulse conditions at the rate of one pulse per minute (max) for ten pulses each, in accordance with 4.5.3 as specified.

4.5.3 Maximum peak pulse current (I_{PP}). The peak pulse currents specified in column 7 of table II shall be applied simultaneously maintaining a bias voltage, not less than the applicable voltage in column 4 of table II, in the same polarity as the peak pulse current. The clamping voltage (V_C) shall be as specified in 4.5.4. The peak pulse current shall be applied with a current versus time waveform (1 pulse per minute maximum) such that the pulse current shall reach 100 percent of I_P at t ≤ 10 µs and decay to 50 percent of I_P at t ≥ 1 ms for t_P = 1 ms (see figure 6).
NOTE: Tolerance on time (t) shall be +10, -0 percent.

4.5.4 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Standby current	4016	Condition A or B; $V_R = V_{WM}$; (column 4 of table II)	I_{D1}		Column 5 of table II	µA dc
Breakdown voltage	4022	$t_P \leq 300$ ms, duty cycle ≤ 2 percent $I_{(BR)1}$ = column 3 of table II	$V_{(BR)1}$	Column 2 of table II		V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Standby current	4016	Condition A or B, $V_R = V_{WM}$ (column 4 of table II)	I_{D2}		Column 9 of table II	µA dc
Subgroup 4						
Clamping voltage maximum (pulsed)		See 4.5.4, t_P = 1 ms (see 4.5.3), I_P = column 7 of table II	$V_{C(max)}$		Column 6 of table II	V (pk)
Subgroups 5						
Not applicable						
Subgroup 6						
Forward surge current	4066	Condition A, one pulse, half-sine wave (8.3 ms), $I_F = 0$; $V_{WM} = 0$; T_A = 25°C				
1N8036 thru 1N8072		I_{FSM} = 25 A (pk)				
1N8073 thru 1N8109		I_{FSM} = 80 A (pk)				
1N8110 thru 1N8146		I_{FSM} = 130 A (pk)				
Subgroup 7						
Forward clamping voltage		See 4.5.4, t_P = 1.0 ms (see 4.5.3) at I_{PP} except use forward direction current without prior bias voltage	V_{CF}		2.5	V(pk)

1/ For sampling plan, see MIL-PRF-19500.
2/ Includes both axial and US packages.

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TABLE II. Electrical characteristics and ratings (for 150 W, 500 W and 1,500 W series)

Column 1			Column 2	Column 3	Column 4	Column 5			Column 6	Column 7			Column 8	Column 9			Column 10		
Series type			$V_{(BR)1}$ Breakdown voltage at $I_{(BR)1}$	$I_{(BR)1}$ Test current	V_{WM} Working peak voltage	I_{D1} Maximum standby current			$V_{C(max)}$ Maximum clamping voltage at I_{PP}	I_{PP} Maximum peak pulse current t_P = 1 ms			$\alpha_{V(BR)}$ Maximum temp. Coeff. of $V_{(BR)}$	I_{D2} Maximum standby current at T_A = +125°C			$I_{(BR)2}$ Breakdown current		
150 W	500 W	1,500 W	Min V 1/	mA 1/	V 1/	µA 2/	µA 3/	µA 4/	V (pk) 1/	A (pk) 2/	A (pk) 3/	A (pk) 4/	% / °C 1/	µA 2/	µA 3/	µA 4/	mA 2/	mA 3/	mA 4/
1N8036	1N8073	1N8110	5.60	10	5.0	500	1000	2000	9.5	15.7	52.6	157	.045	1000	2,000	4,000	90	175	175
1N8037	1N8074	1N8111	6.46	10	5.8	100	300	1000	10.8	13.8	46.2	138	.057	250	750	2000	90	175	175
1N8038	1N8075	1N8112	7.13	10	6.4	20	50	300	11.3	13.2	44.2	132	.061	100	250	1500	90	175	175
1N8039	1N8076	1N8113	7.79	10	6.9	10	20	100	12.1	12.4	41.3	124	.065	60	150	600	75	150	150
1N8040	1N8077	1N8114	8.65	1	7.8	10	20	100	13.4	11.2	37.3	112	.068	60	150	600	75	150	150
1N8041	1N8078	1N8115	9.50	1	8.5	5	20	50	14.5	10.3	34.5	103	.073	20	100	200	65	125	125
1N8042	1N8079	1N8116	10.4	1	9.0	1	5	20	15.6	9.62	32.0	96.2	.075	20	100	200	65	125	125
1N8043	1N8080	1N8117	11.4	1	10.0	1	5	20	16.9	8.88	29.6	88.8	.078	20	100	200	50	100	100
1N8044	1N8081	1N8118	12.4	1	11.0	1	5	20	18.2	8.24	27.5	82.4	.081	20	100	200	50	100	100
1N8045	1N8082	1N8119	13.8	1	12.0	1	5	20	20.2	7.42	24.7	74.2	.084	20	100	200	38	75	75
1N8046	1N8083	1N8120	15.2	1	13.0	0.5	1	5	22.3	6.73	22.4	67.3	.086	20	100	200	38	75	75
1N8047	1N8084	1N8121	17.1	1	15.0	0.5	1	5	25.1	5.98	19.9	59.8	.088	20	100	200	33	65	65
1N8048	1N8085	1N8122	19.0	1	17.0	0.5	1	5	27.7	5.42	18.0	54.2	.090	20	100	200	33	65	65
1N8049	1N8086	1N8123	20.9	1	18.0	0.5	1	5	30.5	4.92	16.4	49.2	.092	20	100	200	25	50	50
1N8050	1N8087	1N8124	22.8	1	20.0	0.5	1	5	33.3	4.50	15.0	45.0	.094	20	100	200	25	50	50
1N8051	1N8088	1N8125	25.7	1	22.0	0.5	1	5	37.4	4.01	13.4	40.1	.096	20	100	200	25	50	50
1N8052	1N8089	1N8126	28.5	1	25.0	0.5	1	5	41.6	3.60	12.0	36.0	.097	20	100	200	20	40	40
1N8053	1N8090	1N8127	31.4	1	28.0	0.5	1	5	45.7	3.28	10.9	32.8	.098	20	100	200	20	40	40
1N8054	1N8091	1N8128	34.2	1	30.0	0.5	1	5	49.9	3.01	10.0	30.1	.099	20	100	200	15	30	30
1N8055	1N8092	1N8129	37.1	1	33.0	0.5	1	5	53.6	2.80	9.3	28.0	.100	20	100	200	15	30	30
1N8056	1N8093	1N8130	40.9	1	36.0	0.5	1	5	59.1	2.54	8.5	25.4	.101	20	100	200	15	30	30
1N8057	1N8094	1N8131	44.7	1	40.0	0.5	1	5	64.6	2.32	7.7	23.2	.101	20	100	200	13	25	25

See footnotes at end of table.

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TABLE II. Electrical characteristics and ratings (for 150 W, 500 W and 1,500 W series) - Continued.

Column 1			Column 2	Column 3	Column 4	Column 5			Column 6	Column 7			Column 8	Column 9			Column 10		
Series type			$V_{(BR)1}$ Breakdown voltage at $I_{(BR)1}$	$I_{(BR)1}$ Test current	V_{WM} Working peak voltage	I_{D1} Maximum standby current			$V_{C(max)}$ Maximum clamping voltage at I_{PP}	I_{PP} Maximum peak pulse current t_P = 1 ms			$\alpha_{V(BR)}$ Maximum temp. Coeff. of $V_{(BR)}$	I_{D2} Maximum standby current at T_A = +125°C			$I_{(BR)2}$ Breakdown current		
150 W	500 W	1,500 W	Min V 1/	mA 1/	V 1/	µA 2/	µA 3/	µA 4/	V (pk) 1/	A (pk) 2/	A (pk) 3/	A (pk) 4/	% / °C 1/	µA 2/	µA 3/	µA 4/	mA 2/	mA 3/	mA 4/
1N8058	1N8095	1N8132	48.5	1	43.0	0.5	1	5	70.1	2.14	7.1	21.4	.102	20	100	200	13	25	25
1N8059	1N8096	1N8133	53.2	1	47.0	0.5	1	5	77.0	1.95	6.5	19.5	.103	20	100	200	10	20	20
1N8060	1N8097	1N8134	58.9	1	53.0	0.5	1	5	85.3	1.76	5.9	17.6	.104	20	100	200	10	20	20
1N8061	1N8098	1N8135	64.6	1	58.0	0.5	1	5	93.7	1.60	5.3	16.0	.104	20	100	200	10	20	20
1N8062	1N8099	1N8136	71.3	1	64.0	0.5	1	5	103.0	1.45	4.8	14.5	.105	20	100	200	10	20	20
1N8063	1N8100	1N8137	77.9	1	70.0	0.5	1	5	113.0	1.32	4.4	13.2	.105	20	100	200	8	15	15
1N8064	1N8101	1N8138	86.5	1	75.0	0.5	1	5	125.0	1.20	4.0	12.0	.105	20	100	200	8	15	15
1N8065	1N8102	1N8139	95.0	1	82.0	0.5	1	5	137.0	1.09	3.6	10.9	.106	20	100	200	6	12	12
1N8066	1N8103	1N8140	104.0	1	94.0	0.5	1	5	152.0	0.98	3.2	9.8	.107	20	100	200	6	12	12
1N8067	1N8104	1N8141	114.0	1	100.0	0.5	1	5	168.0	0.89	2.9	8.9	.107	20	100	200	5	10	10
1N8068	1N8105	1N8142	124.0	1	110.0	0.5	1	5	183.0	0.82	2.7	8.2	.107	20	100	200	5	10	10
1N8069	1N8106	1N8143	138.0	1	120.0	0.5	1	5	208.0	0.72	2.4	7.2	.108	20	100	200	4	8	8
1N8070	1N8107	1N8144	148.0	1	130.0	0.5	1	5	225.0	0.67	2.2	6.7	.108	20	100	200	4	8	8
1N8071	1N8108	1N8145	171.0	1	150.0	0.5	1	5	261.0	0.57	1.9	5.7	.108	20	100	200	3	5	5
1N8072	1N8109	1N8146	190.0	1	170.0	0.5	1	5	294.0	0.51	1.7	5.1	.108	20	100	200	3	5	5

1/ Applies to 150 W, 500 W and 1,500 W series.
2/ Applies to only 150 W series
3/ Applies to only 500 W series.
4/ Applies to only 1,500 W series.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			n = 45 devices, c = 0
Temperature cycling	1051	500 cycles, condition C, -55°C to +175°C.	
Electrical measurements		See table I, subgroup 2.	
Subgroup 2			n = 45 devices, c = 0
Life test		1,000 hours. See 4.5.1.	
Electrical measurements		See table I, subgroup 2.	
Subgroups 4, 5, and 6			
Not applicable			
Subgroup 7			n = 3 devices, c = 0
Soldering heat	2031	Condition A and (B or C).	
Subgroup 8			n = 45 devices, c = 0
Peak pulse current		See 4.5.3. I_{PP} shall be characterized by the supplier and this data shall be available to the Government. Test shall be performed on each low and high voltage device for each structurally identical grouping. Test to failure. (See figures 7 and 8).	
Electrical measurements		See table I, subgroup 2.	

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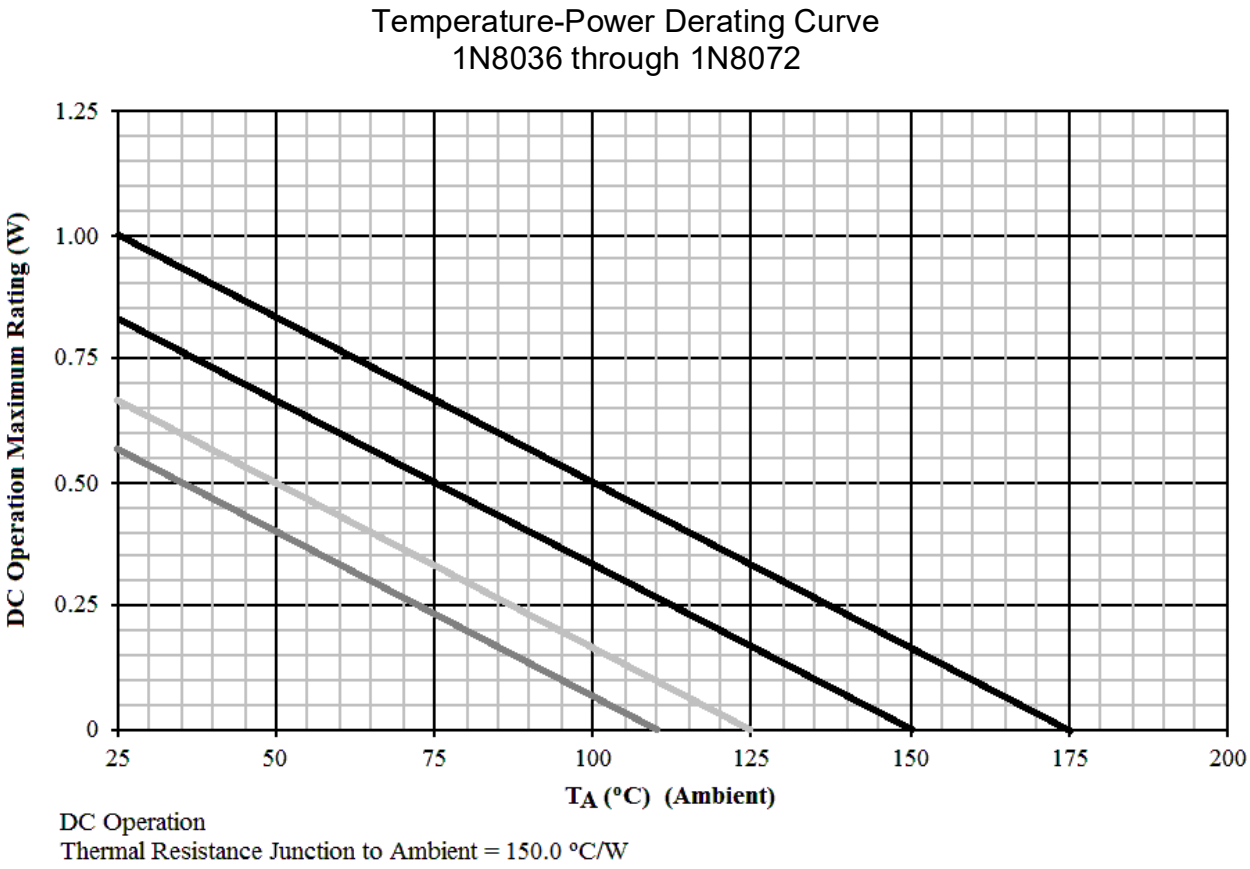


FIGURE 3. Maximum dc power derating versus lead temperature.

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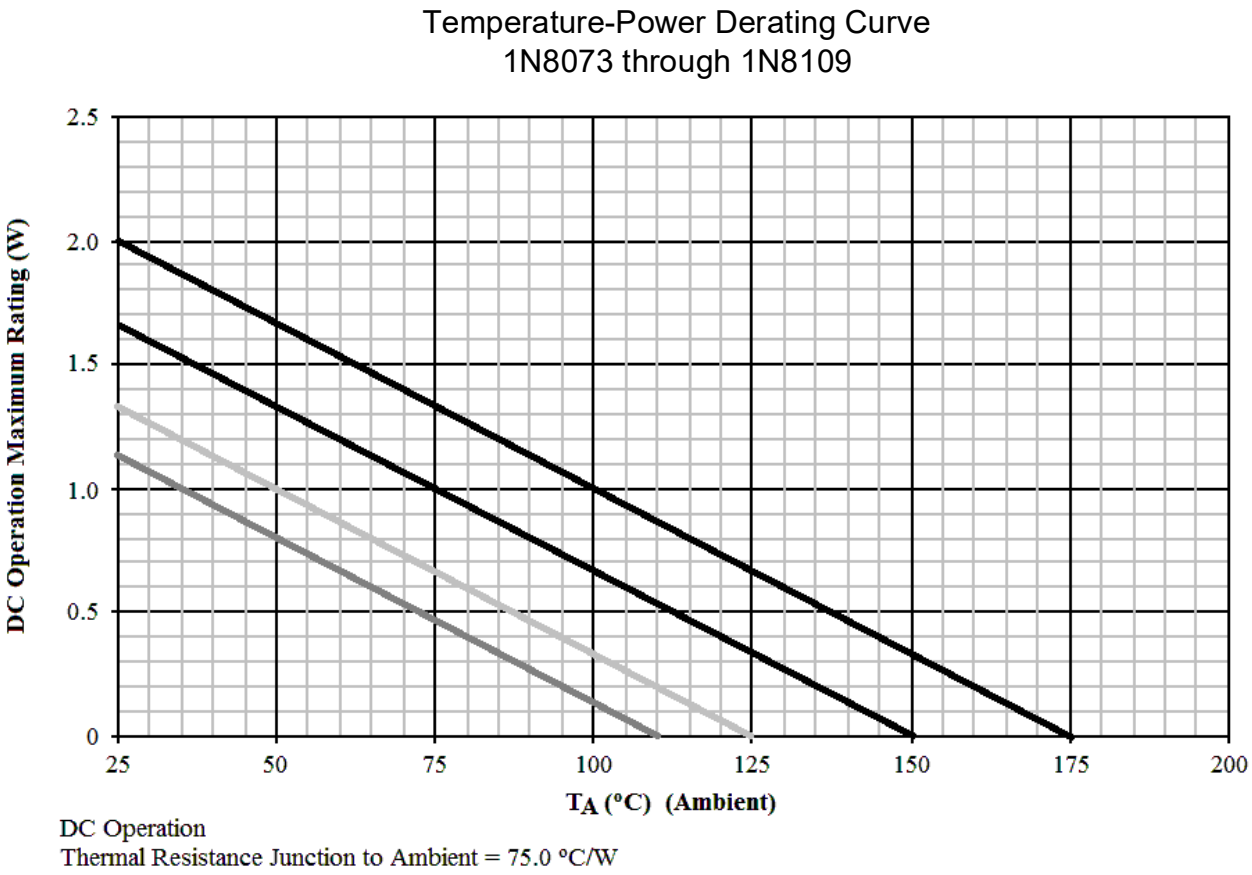


FIGURE 4. Maximum dc power derating versus lead temperature.

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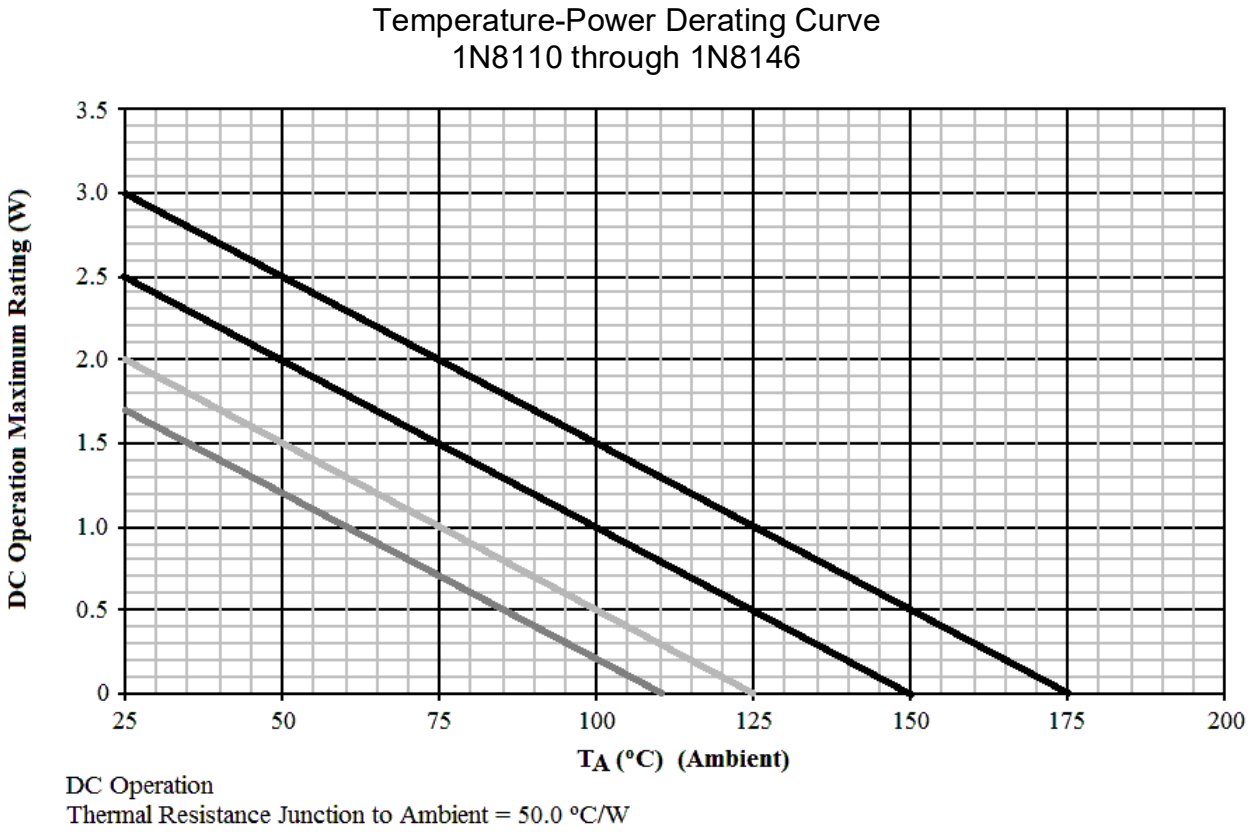


FIGURE 5. Maximum dc power derating versus lead temperature.

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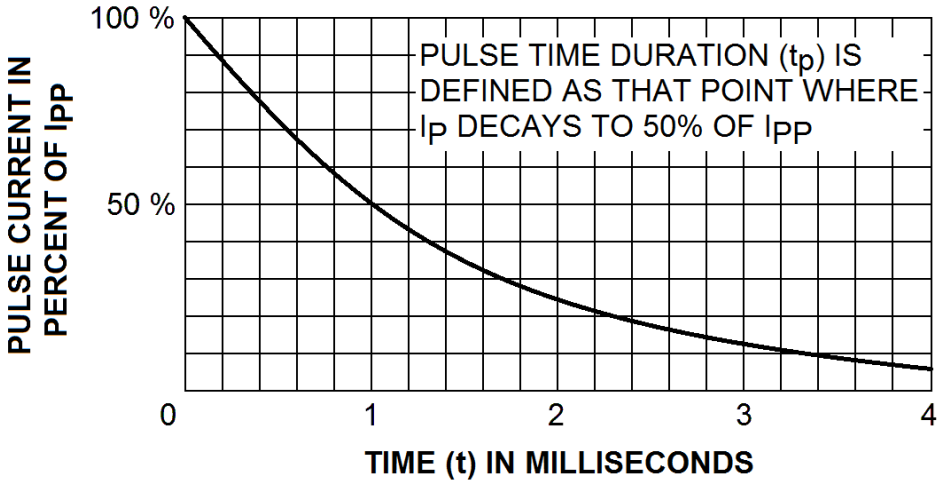


FIGURE 6. Pulse waveform.

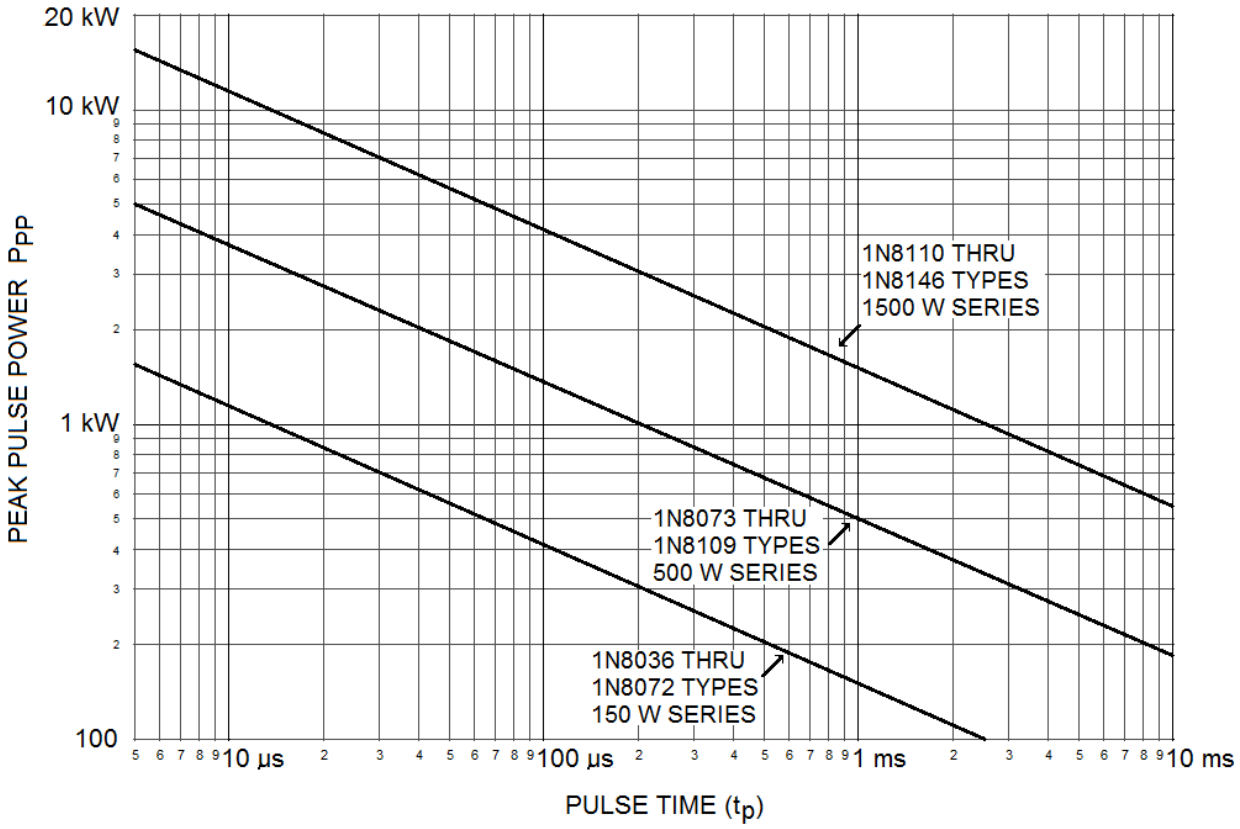
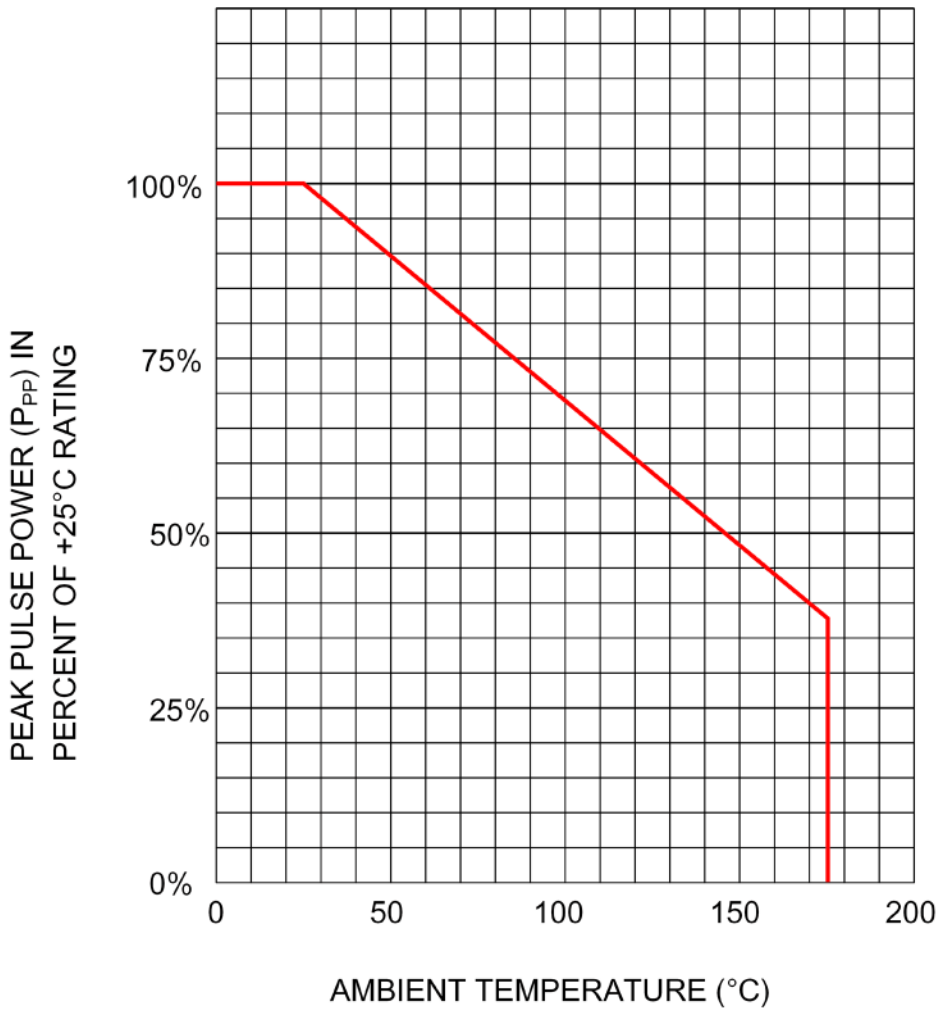


FIGURE 7. Peak pulse power versus pulse time.

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NOTES:
1. Pulse condition defined in figure 7.
2. The pulse derating curve of maximum peak pulse power versus junction temperature has been included for reference purposes only.

FIGURE 8. Pulse derating curve.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

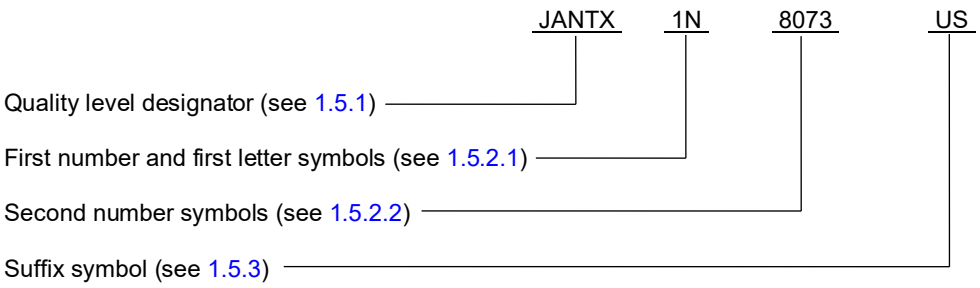
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

e. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2023-039)

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